XIANG LI

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EDUCATION

Peking University, Beijing, China Sep. 2018 – Jul. 2023
Ph.D. in Statistics, School of Mathematical Sciences
Advisor: Prof. Zhihua Zhang

Peking University, Beijing, China Sep. 2014 – Jul. 2018
B.S. in Statistics, School of Mathematical Sciences
B.A. in Economics, National School of Development

EXPERIENCE

University of Pennsylvania, Philadelphia, USA Aug. 2023 – Present
Postdoctoral Researcher, Department of Statistics and Data Science
Working with Prof. Weijie Su and Prof. Qi Long

University of Toronto, Toronto, Canada Feb. 2022 – Feb. 2023
Visiting Ph.D. Student, Department of Statistical Sciences
Working with Prof. Qiang Sun

Face++ (Megvii), Beijing, China Jun. 2018 – Nov. 2018
Research Intern, Algorithm Group
Mentor: Dr. Shuchang Zhou

ToSimple, Beijing, China Mar. 2018 – May. 2018
Research Intern, Algorithm Group
Mentor: Dr. Naiyan Wang

SELECTED AWARDS

- IMS New Researcher Travel Award 2025
- Outstanding PhD Graduates, Peking University 2023
- President Scholarship for Excellent Ph.D Student, Peking University 2018, 2020-2021
- Outstanding reviewer in NeurIPS, top 8% 2021
- National Scholarship, China 2017, 2019
- Travel Award NeurIPS 2019, AAAI 2020
- Outstanding Graduates, School of Mathematical Sciences, Peking University 2018
- First prize in National Undergraduate Mathematical Modeling Contest, China 2016

RESEARCH INTERESTS

My research lies at the intersection of statistics, optimization, and machine learning. Currently, I focus on principled and scalable methodologies for generative AI—particularly large language models—with the goal of advancing theoretical understanding and developing insights that support safe, robust, and reliable deployment. Specifically, my research spans the following directions:

- **Statistical Foundations of Generative AI**

- **Provenance and Safety Mechanisms.** Designing watermarking schemes [J4, S2, S3] and developing detection [J1, J2, C3, W1] and estimation [J3] methods for tracing and verifying AI outputs, with the broader aim of building scalable safety mechanisms that safeguard provenance, accountability, and responsible deployment.
- **Fundamental Principles.** Modeling generative AI systems as black boxes and developing statistical tools to analyze input–output relationships, focusing on core issues such as evaluation [S1], uncertainty quantification [S4], scaling law, privacy preserving, and model collapse.

- **Reliable Statistical Machine Learning**
  - **Learning under Challenging Conditions.** Developing efficient algorithms resilient to nonstandard conditions such as heavy-tailed noise [J8], corruption [C5], high-dimensionality [S5], and data heterogeneity, and analyzing their statistical or convergence properties [C12, C7].
  - **Uncertainty Quantification in Adaptive Environments.** Designing statistical methods for streaming data [J5, S7] and adaptive decision-making [C6, C11, C13], ensuring valid statistical inference in dynamic or intricate environments [C2, S8, J6, S6].
- **Towards Efficient Federated Learning**
  - **Communication Efficiency.** Developing communication-efficient algorithms [J9, C10, T2] and analyzing their convergence properties under heterogeneous data settings [C1, C8, T1].
  - **Trustworthiness.** Exploring key aspects of trustworthy federated learning, including differential privacy [C4, S9] and personalization [J7, C9].

PUBLICATIONS

* indicates equal contribution. ** indicates alphabetical order.

Journal articles (Published or under revision)

[J1] **A Statistical Framework of Watermarks for Large Language Models: Pivot, Detection Efficiency and Optimal Rules** [pdf] [arxiv]
Xiang Li, Feng Ruan, Huiyuan Wang, Qi Long, Weijie J. Su
Annals of Statistics, 53 (1): 322-351, Feb. 2025
Selected for presentation at the Annals of Statistics Invited Paper Session, JSM 2025.

[J2] **Robust Detection of Watermarks for Large Language Models Under Human Edits** [arxiv]
Xiang Li, Feng Ruan, Huiyuan Wang, Qi Long, Weijie J. Su
To appear in *Journal of the Royal Statistical Society Series B*
Received the 2025 IMS New Researcher Travel Award.

[J3] **Optimal Estimation of Watermark Proportions in Hybrid AI-Human Texts** [arxiv]
Xiang Li, Garrett Wen, Weiqing He, Jiayuan Wu, Qi Long, Weijie J. Su
Major revision at *Journal of the American Statistical Association*.

[J4] **Debiasing Watermarks for Large Language Models via Maximal Coupling** [pdf] [arxiv]
Yangxinyu Xie, **Xiang Li**, Tanwi Mallick, Weijie J. Su, Ruixun Zhang
Journal of the American Statistical Association, 1–11, 2025

[J5] **Convergence and Inference of Stream SGD, with Applications to Queueing Systems and Inventory Control** [arxiv]
Xiang Li*, Jiadong Liang*, Xinyun Chen, Zhihua Zhang
Minor revision at *Operation Research*

[J6] **Decoupled Functional Central Limit Theorems for Two-Time-Scale Stochastic Approximation** [arxiv]
Yuze Han, **Xiang Li**, Jiadong Liang, Zhihua Zhang
Major revision at *Mathematics of Operations Research*

[J7] **A Random Projection Approach to Personalized Federated Learning: Enhancing Communication Efficiency, Robustness, and Fairness** [pdf]
Yuze Han**, **Xiang Li****, Shiyun Lin**, Zhihua Zhang**
Journal of Machine Learning Research, 25 (380): 1-88, Dec. 2024

[J8] **Variance-aware Decision Making with Linear Function Approximation with Heavy-tailed Rewards** [pdf] [arxiv]
Xiang Li, Qiang Sun
Transactions on Machine Learning Research, Apr. 2024
Invited to present at ICLR 2025; ranked among the Top 100 papers accepted by TMLR that year.

[J9] **FedPower: Privacy-Preserving Distributed Eigenspace Estimation** [pdf] [arxiv]
Xiao Guo*, **Xiang Li***, Xiangyu Chang, Shusen Wang, Zhihua Zhang
Machine Learning, 113 (11): 8427-8458, Sep. 2024

Conference papers

[C1] **On the Convergence of FedAvg on Non-IID Data** [pdf]
Xiang Li*, Kaixuan Huang*, Wenhao Yang*, Shusen Wang, Zhihua Zhang
International Conference on Learning Representations (ICLR) 2020 (Oral presentation, Google citation 3300+)

[C2] **Statistical Estimation and Online Inference via Local SGD** [pdf]
Xiang Li, Jiadong Liang, Xiangyu Chang, Zhihua Zhang
Conference on Learning Theory (COLT) 2022

[C3] **On the Empirical Power of Goodness-of-Fit Tests in Watermark Detection** [arxiv]
Weiqing He* and **Xiang Li***, Tianqi Shang, Li Shen, Weijie J. Su, Qi Long
Neural Information Processing Systems (NeurIPS) 2025 (Spotlight)

[C4] **Mitigating the Privacy–Utility Trade-off in Decentralized Federated Learning via f-Differential Privacy**
Xiang Li, Chendi Wang, Buxin Su, Qi Long, Weijie J. Su
Neural Information Processing Systems (NeurIPS) 2025 (Spotlight)

[C5] **Corruption-Robust Variance-aware Algorithms for Generalized Linear Bandits under Heavy-tailed Rewards** [pdf]
Qingyuan Yu, Euijin Baek, **Xiang Li**, Qiang Sun
Conference on Uncertainty in Artificial Intelligence (UAI) 2025

[C6] **A Statistical Analysis of Polyak-Ruppert-Averaged Q-Learning** [pdf]
Xiang Li, Wenhao Yang, Jiadong Liang, Zhihua Zhang, Michael I. Jordan
International Conference on Artificial Intelligence and Statistics (AISTATS) 2023

[C7] **Statistical Analysis of Karcher Means for Random Restricted PSD Matrices** [pdf]
Hengchao Chen, **Xiang Li**, Qiang Sun
International Conference on Artificial Intelligence and Statistics (AISTATS) 2023

[C8] **Asymptotic Behaviors of Projected Stochastic Approximation: A Jump Diffusion Perspective** [pdf]
Jiadong Liang, Yuze Han, **Xiang Li**, Zhihua Zhang
Neural Information Processing Systems (NeurIPS) 2022 (Spotlight)

[C9] **Personalized Federated Learning towards Communication Efficiency, Robustness and Fairness** [pdf]
Shiyun Lin*, Yuze Han*, **Xiang Li**, Zhihua Zhang
Neural Information Processing Systems (NeurIPS) 2022

[C10] **Communication-Efficient Distributed SVD via Local Power Iterations** [pdf]
Xiang Li, Shusen Wang, Kun Chen, Zhihua Zhang
International Conference on Machine Learning (ICML) 2021

[C11] **Finding the Near Optimal Policy via Reductive Regularization in MDPs** [pdf]
Wenhao Yang, **Xiang Li**, Guangzeng Xie, Zhihua Zhang
Workshop on Reinforcement Learning Theory, ICML 2021

[C12] **Do Subsampled Newton Methods Work for High-Dimensional Data?** [pdf]
Xiang Li, Shusen Wang, Zhihua Zhang
AAAI Conference on Artificial Intelligence (AAAI) 2020

[C13] **A Regularized Approach to Sparse Optimal Policy in Reinforcement Learning** [pdf]
Wenhao Yang*, **Xiang Li***, Zhihua Zhang
Neural Information Processing Systems (NeurIPS) 2019

Submitted (under review)

[S1] **Evaluating the Unseen Capabilities: How Many Theorems Do LLMs Know?** [arxiv]
Xiang Li, Jiayi Xin, Qi Long, Weijie J. Su

[S2] **Break the Trade-off Between Watermark Strength and Speculative Sampling Efficiency for Language Models**
Weiqing He*, **Xiang Li***, Li Shen, Weijie J. Su, Qi Long

[S3] **TAB-DRW: A DFT-based Robust Watermark for Generative Tabular Data**
Yizhou Zhao, **Xiang Li**, Peter Song, Qi Long, Weijie J. Su

[S4] **Paraphrase-Robust Conformal Prediction for Reliable LLM Uncertainty Quantification**
Jiayi Xin, Evan Qiang, **Xiang Li**, Weijie J. Su, Qi Long

[S5] **High-Dimensional Nonparametric Density Estimation via Max-Random Forest**
Xiang Li, Yiliang Zhang, Jiayuan Wu, Qi Long, Weijie J. Su

[S6] **Uncertainty Quantification of Data Shapley via Statistical Inference** [arxiv]
Mengmeng Wu, Zhihong Liu, **Xiang Li**, Ruoxi Jia, Xiangyu Chang

[S7] **Finite-Time Decoupled Convergence in Nonlinear Two-Time-Scale Stochastic Approximation** [arxiv]
Yuze Han**, **Xiang Li****, Zhihua Zhang**

[S8] **Online Statistical Inference for Nonlinear Stochastic Approximation with Markovian Data** [arxiv]
Xiang Li, Jiadong Liang, Zhihua Zhang

[S9] **Privacy-Preserving Community Detection for Locally Distributed Multiple Networks** [arxiv]
Xiao Guo, **Xiang Li**, Xiangyu Chang, Shujie Ma

Working papers (Manuscript available upon request)

[W1] **Optimal Detection for Language Watermarks with Pseudorandom Collision**
T. Tony Cai**, **Xiang Li****, Qi Long**, Weijie J. Su**, Garrett G. Wen**

Technical reports

[T1] **Asymptotic Behaviors and Phase Transitions in Projected Stochastic Approximation: A Jump Diffusion Approach** [arxiv]
Jiadong Liang, Yuze Han, **Xiang Li**, Zhihua Zhang

[T2] **Communication-Efficient Local Decentralized SGD Methods** [arxiv]
Xiang Li, Wenhao Yang, Shusen Wang, Zhihua Zhang

GRANTS & FUNDING

- Contributor to NIH RO1 Grant Proposal, co-written with Prof. Weijie J. Su & Prof. Qi Long (2024)
  - Assisted in proposal writing, structuring key research points on watermarking.

PRESENTATIONS

- *Evaluating the Unseen Capabilities: How Many Theorems Do LLMs Know?*
  - Sep. 2025, AI Research Mixer 2025, UPenn.
- *Robust Watermark Detection and Efficient Proportion Estimation for Human-Edited Watermarked Text*
  - Jun. 2025, ICSA Applied Statistics Symposium, University of Connecticut.
  - Aug. 2025, Joint Statistical Meetings, Nashville, Tennessee.
- *Optimal Robust Detection for Gumbel-Max Watermarks Under Contamination*
  - Jul. 2024, IMS-China International Conference on Statistics and Probability, Yinchuan.
  - Sep. 2024, Penn conference on Big Data in Biomedical & Population Health Sciences.
  - Sep. 2024, Warren & ASSET Center Research Mixer, UPenn.
- *A Statistical Framework of Watermarks for Large Language Models: Pivot, Detection Efficiency and Optimal Rules*
  - May 2024, the NUS IMS workshop on statistical machine learning for high dimensional data, Singapore.
  - Jul. 2024, the 2nd joint conference on statistics and data science in China (JCSDS), Kunming.
  - Nov. 2024, the conference on statistical learning and data science (SLDS), Newport Beach, California.
- *Complete Asymptotic Analysis for Projected Stochastic Approximation and Debiased Variants*
  - Sep. 2023, the Allerton conference at the University of Illinois, Allerton Park & Retreat Center.
- *Polyak-Ruppert-Averaged Q-Learning is Statistically Efficient*
  - Jul. 2022, the RL workshop in Shanghai University of Finance and Economics.
- *Statistical Estimation and Inference via Local SGD in Federated Learning*
  - Nov. 2021, the 14-th China-R conference.
- *A Regularized Approach to Sparse Optimal Policy in Reinforcement Learning*
  - Jun. 2019, the machine learning workshop at Peking University.

TEACHING EXPERIENCES

Instructor

- Short course (*Statistical Inference in Large Language Models*) at the 2025 ICSA Applied Statistics Symposium [slides] Summer 2025

Invited guest lecturer

- STAT 9917: *Statistical Topics in Large Language Models*, UPenn Spring 2025

Teaching assistant

- *Linear Algebra*, Peking University Spring 2021
- *Reinforcement Learning: Theory and Algorithms*, Peking University Fall 2019

ACADEMIC SERVICE

Conference review

- NeurIPS, ICML, ICLR, UAI, AISTATS, IJCAI.

Journal review

- Journal of Machine Learning Research
- Transactions on Machine Learning Research
- Journal of the American Statistical Association
- Information and Inference: A Journal of the IMA
- Annals of Applied Probability
- Operational Research
- IEEE Transactions on Automatic Control
- IEEE Open Journal of Signal Processing
- IEEE Journal on Selected Areas in Communications
- Transactions on Parallel and Distributed Systems
- World Wide Web

Event organization

- Aug. 2024, Session Chair for "Theoretical and Algorithmic Developments in Federated Learning" at the Modeling and Optimization: Theory and Applications (MOPTA) conference.

MENTORING

- Yuze Han [J6, J7, C9, S7] PhD student at PKU → Lecturer at Renmin Univ. of China
- Jiadong Liang [J5, C2, C6, C8, S8] PhD student at PKU → Postdoc at UPenn
- Yangxinyu Xie [J4] PhD student at UPenn
- Garrett G. Wen [J3, W1] PhD student at Yale Univ.
- Weiqing He [C3, S2] PhD student at UPenn
- Jiayi Xin [S1, S4] PhD student at UPenn
- Yizhou Zhao [S3] Master's student at UPenn

LANGUAGES

- Chinese Mandarin: Native
- English: Fluent

REFEREES

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Daniel H. Silberberg Professor
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The Wharton School, University of Pennsylvania
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tcai@wharton.upenn.edu

Professor Weijie J. Su
Associate Professor
Department of Statistics and Data Science
The Wharton School, University of Pennsylvania
Address: 411 Academic Research Building, 265 South 37th Street, Philadelphia, PA 19104
suw@wharton.upenn.edu

Professor Qi Long
Professor and Vice Chair
Department of Biostatistics, Epidemiology and Informatics, Perelman School of Medicine
University of Pennsylvania
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Professor Zhihua Zhang
Professor
Department of Probability and Statistics
School of Mathematical Sciences, Peking University
Address: Science building No. 1, 5 Yiheyuan Road, Beijing 100871, China
zhzhang@math.pku.edu.cn